

Assessing Causality

Definition: Causality assessment is the method by which the extent of the relationship between a drug and a suspected adverse reaction is estimated.

Methods of Causality Assessment

Group-1	Group-2
Opinion of Experts, Clinical Judgment, or Global Introspection methods	Algorithms (with or without scoring) or Standardised Assessment methods

Group-1: WHO-UMC Causality Assessment Scale

Causation is established based on the clinical judgment of the expert or panel of experts. Such judgments are mostly based on the **WHO-UMC causality assessment scale** given below:

Causality Term	Assessment Criteria*
Certain	• Event or laboratory test abnormality, with plausible time relationship to drug intake • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognised pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable / Likely	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear

Causality Term	Assessment Criteria*
Unlikely	• Event or laboratory test abnormality, with a time to drug intake that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations
Conditional / Unclassified	• Event or laboratory test abnormality • More data for proper assessment needed, or additional data under examination
Unassessable / Unclassifiable	• A report suggesting an adverse reaction • Cannot be judged because information is insufficient or contradictory • Data cannot be supplemented or verified

* Assessment criteria based on World Health Organization – Uppsala Monitoring Centre (WHO-UMC) standard definitions.